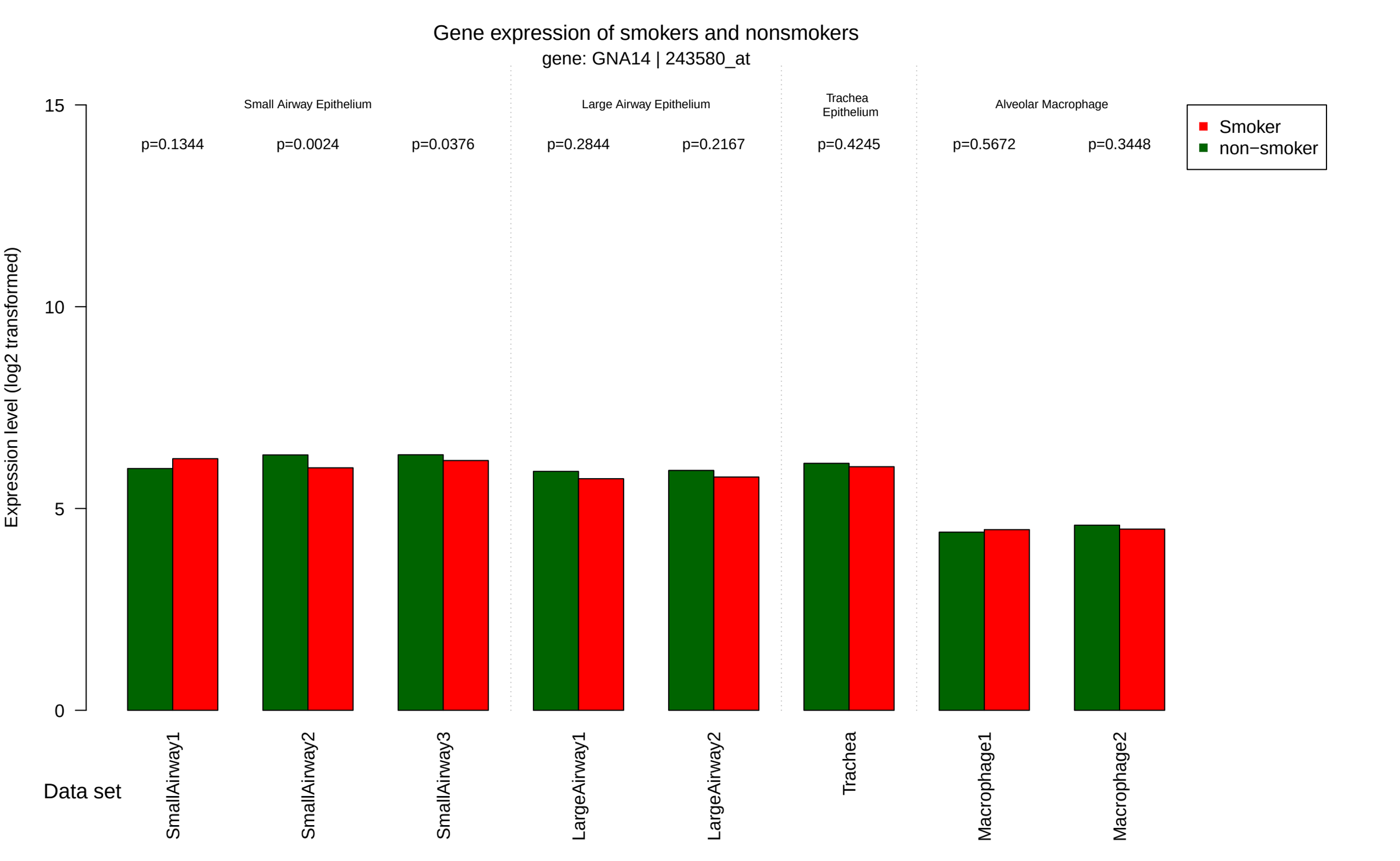
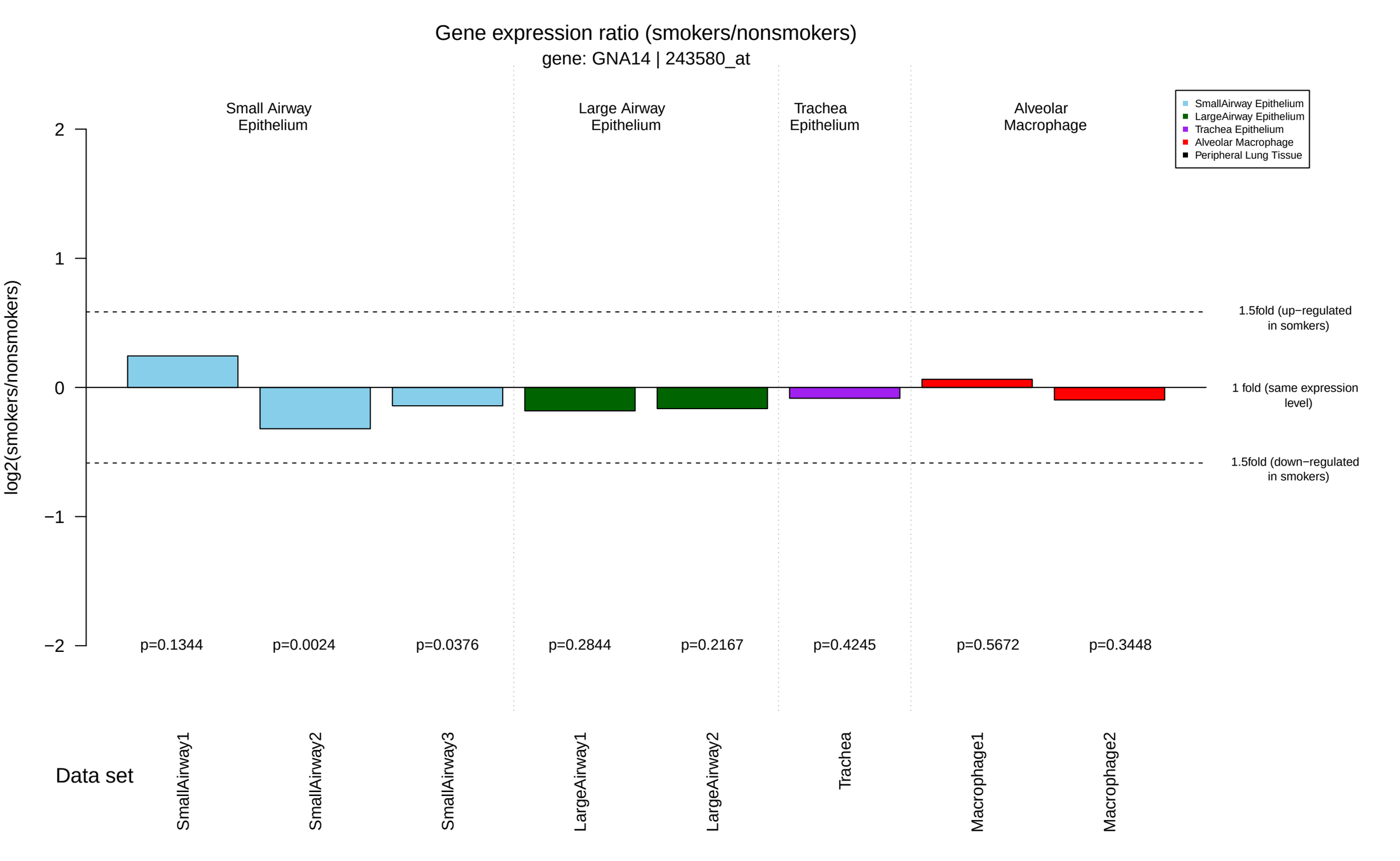
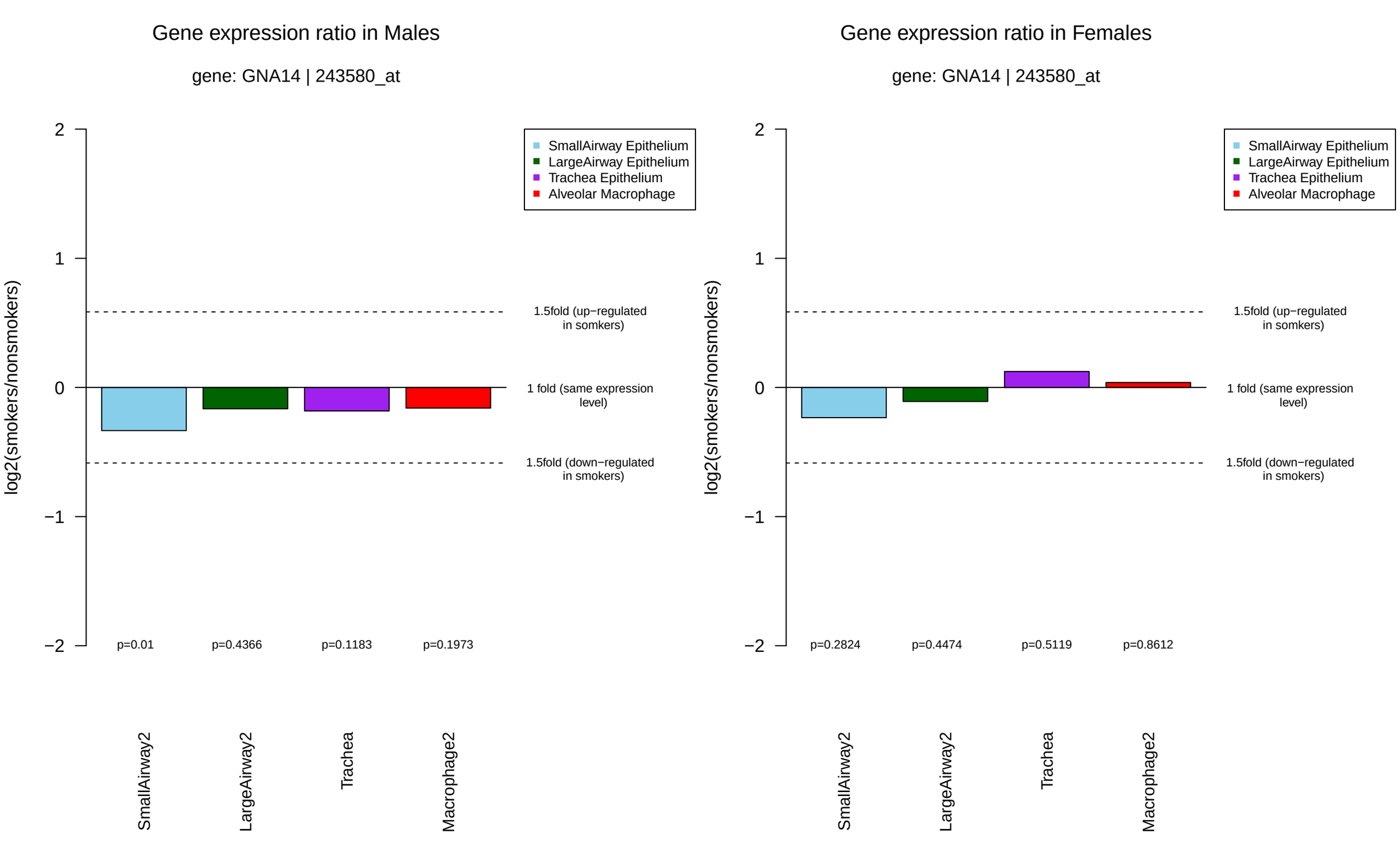
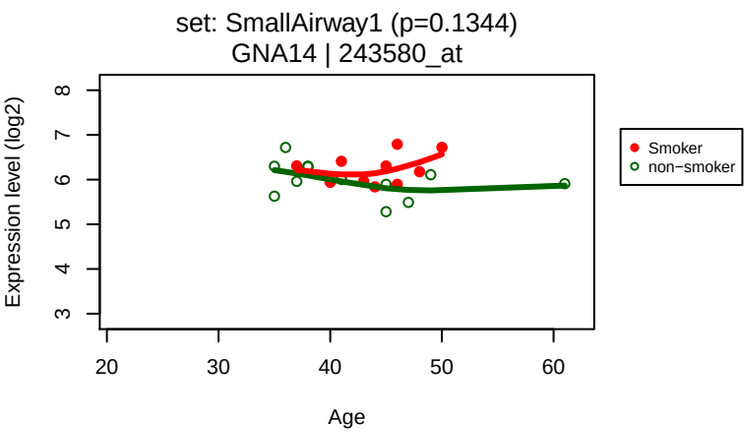
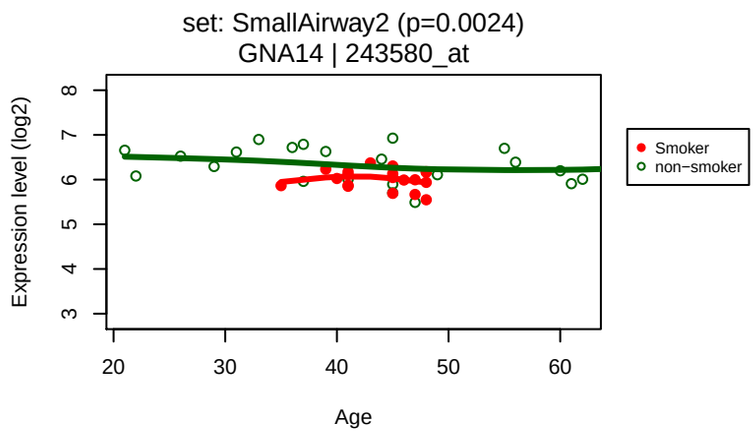
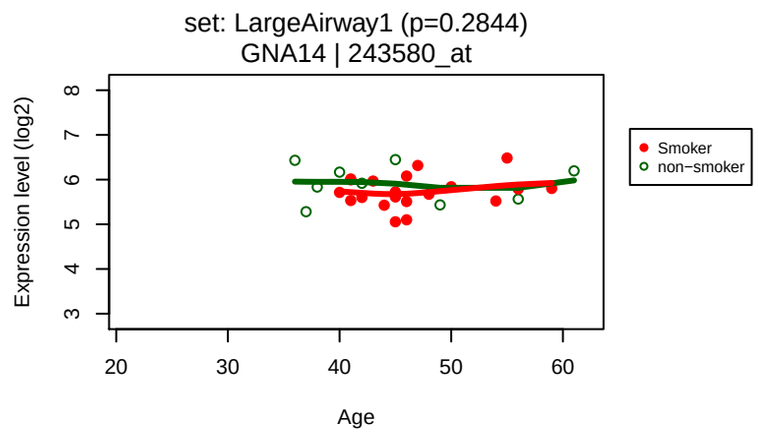

Gene expression of smokers and nonsmokers
gene: GNA14 | 243580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1344
p=0.0024
p=0.0376
p=0.2844
p=0.2167
p=0.4245
p=0.5672
p=0.3448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

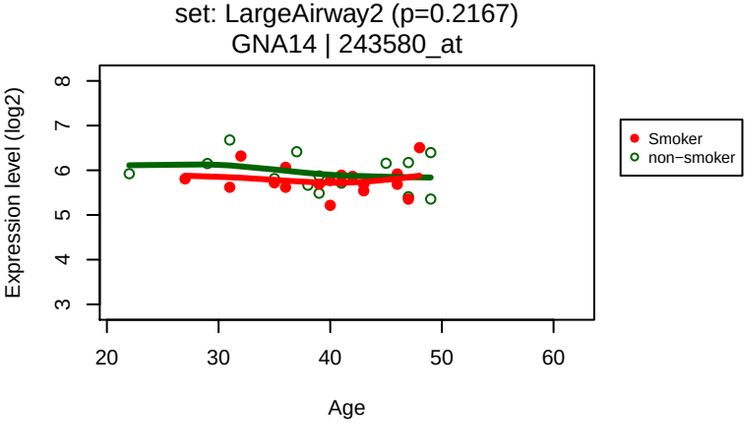

Gene expression ratio (smokers/nonsmokers)
gene: GNA14 | 243580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1344
p=0.0024
p=0.0376
p=0.2844
p=0.2167
p=0.4245
p=0.5672
p=0.3448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

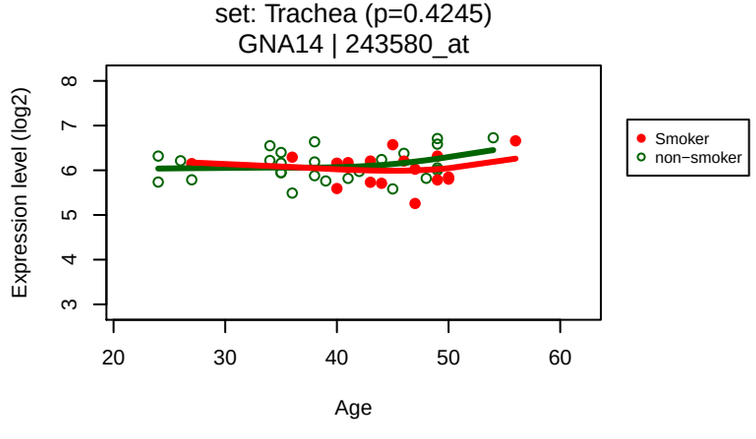

Gene expression ratio in Males
gene: GNA14 | 243580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.01
p=0.4366
p=0.1183
p=0.1973
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA14 | 243580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2824
p=0.4474
p=0.5119
p=0.8612
SmallAirway2
LargeAirway2
Trachea
Macrophage2

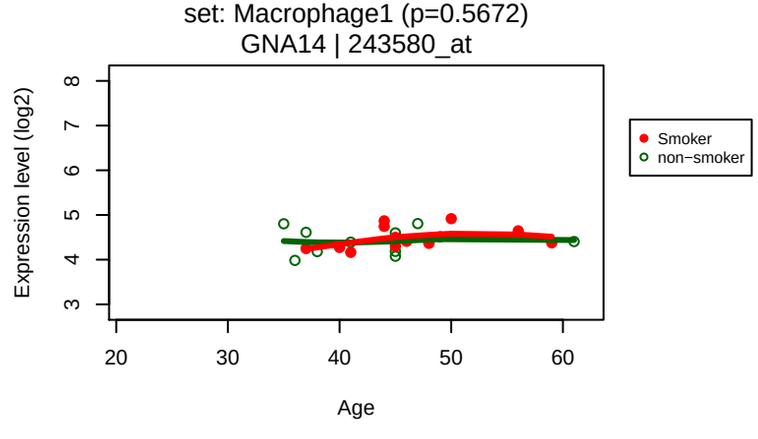

set: SmallAirway1 (p=0.1344)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker

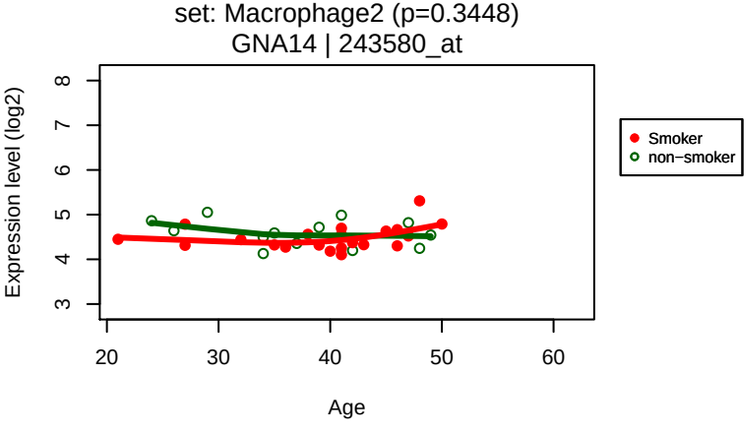

set: SmallAirway2 (p=0.0024)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker

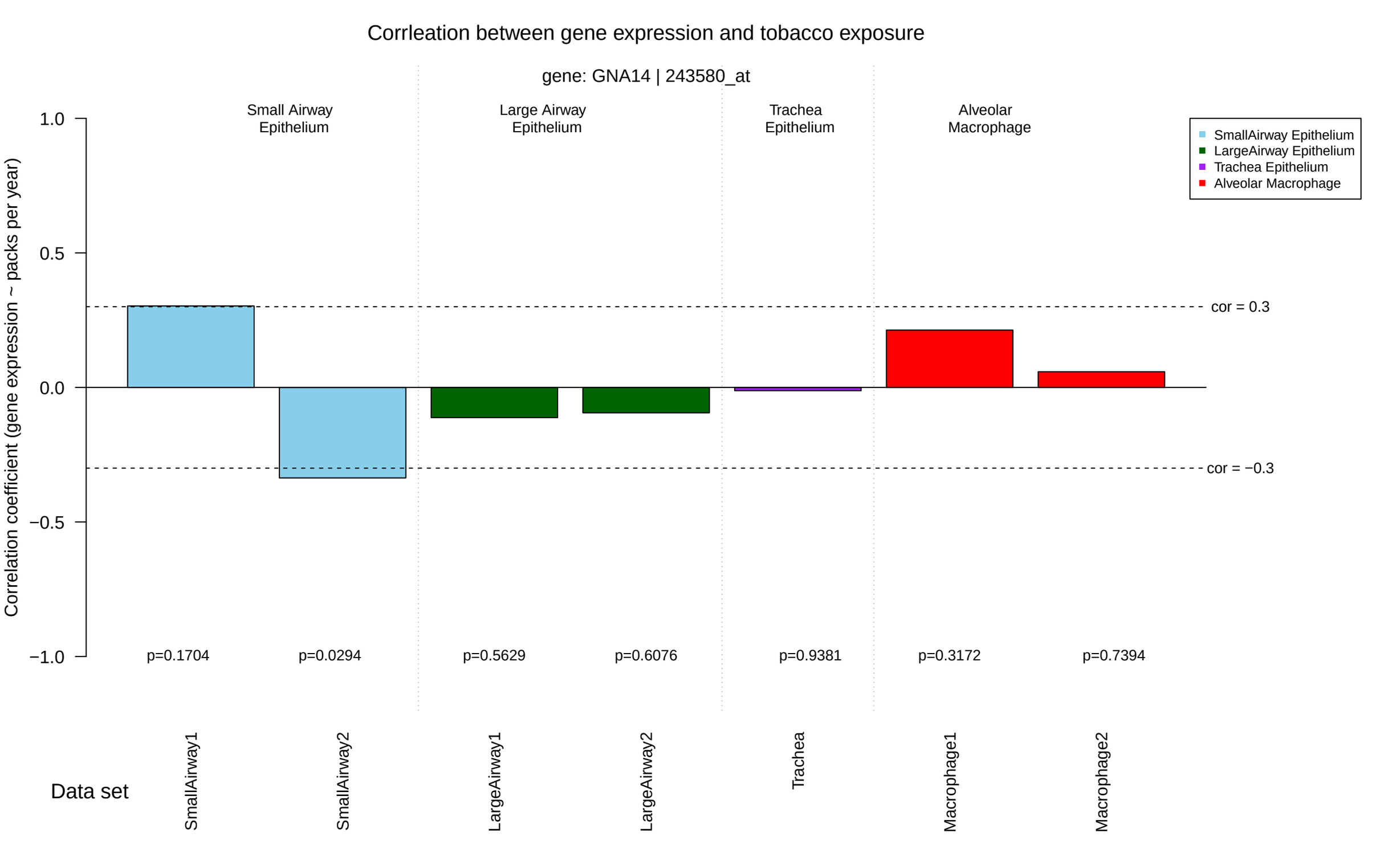

set: LargeAirway1 (p=0.2844)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2167)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4245)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5672)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3448)
GNA14 | 243580_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA14 | 243580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1704
p=0.0294
p=0.5629
p=0.6076
p=0.9381
p=0.3172
p=0.7394
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2